SST Falsifier Report

A063-v0.1.1
Hicks-Hill analytic Euler separatrix controls
External read-only rendering; not an original frozen report.
Generated: 2026-10-13T14:41:40.209948+00:00

Evidence boundary

No new science was executed for this report. Gate outcomes are copied from the source ledger without reclassification. Missing records are not silently filled.

Gate summary

PASS=10, UNRESOLVED=3

Gate	Status	Preregistered question
H0	PASS	Versioned source/protocol bytes match release seal?
H1	PASS	Do analytical roots satisfy registered equations?
H2	PASS	Does Hill zero-swirl control converge under finite-difference refinement?
H3	PASS	Does first Hicks Beltrami field satisfy curl(u)=k*u, Euler and divergence under refinement?
H4	PASS	Does lambda2 first Hicks mode have zero translation and a convergent stationary Euler field?
H5	PASS	Is the spherical boundary a material surface in all three exact fields?
H6	PASS	Do full-latitude interior/exterior pressure and velocity jumps converge to zero?
H7	UNRESOLVED	Are torus-knot accessibility and finite-core vortex-tube stability both certified?
H7A	PASS	Can a first-Beltrami streamline realize a closed rational 3/4 torus-winding candidate?
H7B	UNRESOLVED	Is trefoil T(2,3) accessible as a finite-core stable tube?
H8	UNRESOLVED	Has independent C++ FP64 certification run through v1.0.7 frozen protocol?
H9	PASS	Does exact pressure satisfy incompressible pressure-Poisson closure?
H10	PASS	Does finite laboratory-frame kinetic-energy quadrature converge?

Gate detail (source-derived)

H0 — PASS

Versioned source/protocol bytes match release seal?

Release-side file hash audit; not a canonical v1.0.7 FREEZE.

- protocol_sha256: 838a73226435889fe3cfee63053ee12d6a4a06d7cde67bb74a7ad02296038b99
- source_files_checked: 8
- scope: standalone analytical reference

H1 — PASS

Do analytical roots satisfy registered equations?

Hicks lambda1/lambda2 control equations.

- max_root_equation_abs_residual: 9.547918011776346e-14
- lambda1: array, 3 elements
- lambda2: array, 3 elements

H2 — PASS

Does Hill zero-swirl control converge under finite-difference refinement?

*Finer 5e-5*a finite differences; no stability inference.*

- max_euler: 9.215796285574052e-09
- max_div: 4.3582915054685145e-12
- max_beltrami: 6.075000000000941
- euler_error_ratio_prev_to_finest: 3.9984057649486395
- fd_steps_h_over_a: array, 4 elements

H3 — PASS

Does first Hicks Beltrami field satisfy curl(u)=k*u, Euler and divergence under refinement?

Beltrami-only maxima; other families never pooled into H3.

- max_euler: 8.872942953380721e-07
- max_div: 3.255360248033412e-08
- max_beltrami: 1.705121776902665e-07
- euler_error_ratio_prev_to_finest: 3.9998956136591133
- fd_steps_h_over_a: array, 4 elements

H4 — PASS

Does lambda2 first Hicks mode have zero translation and a convergent stationary Euler field?

Stationary family is not Beltrami; zero translation is not zero internal motion.

- translation_U: 5.921189464667501e-16
- max_euler: 4.262860860724337e-06
- max_div: 1.0834632213629902e-07
- max_beltrami: 25.671855008223712
- euler_error_ratio_prev_to_finest: 3.999446084136766

H5 — PASS

Is the spherical boundary a material surface in all three exact fields?

Spherical material boundary in each steady reference field, not a perturbation result.

- max_surface_normal_velocity: 1.700029006457271e-15
- max_surface_psi: 8.598555549939106e-16
- theta_samples_per_family: 121

H6 — PASS

Do full-latitude interior/exterior pressure and velocity jumps converge to zero?

121 polar angles in each of three families; extrapolating radial limits, not a single-point comparison.

- surface_theta_samples: 121
- radial_offsets: array, 3 elements
- convergence_all: True

H7 — UNRESOLVED

Are torus-knot accessibility and finite-core vortex-tube stability both certified?

A closed streamline candidate is not a certified knotted finite vortex core or nonlinear stability.

- sampled_windings: array, 5 elements
- candidate: T(3,4)_NUMERICAL_CANDIDATE

H7A — PASS

Can a first-Beltrami streamline realize a closed rational 3/4 torus-winding candidate?

Numerical T(3,4) streamline candidate only; no independent knot polynomial or tube stability.

- candidate: T(3,4)_NUMERICAL_CANDIDATE
- s0_over_a: 0.9683098301738062
- full_3d_closure_relative_a: 9.183241962123216e-10
- winding_target_abs_error: 1.857403120197887e-13
- total_azimuthal_winding: 2.9999999999570015

H7B — UNRESOLVED

Is trefoil T(2,3) accessible as a finite-core stable tube?

No global winding enclosure, interval proof, independent knot invariant or finite-core tube perturbation certification.

- trefoil_winding_global_bounds_certified: False
- vortex_tube_stability_certified: False

H8 — UNRESOLVED

Has independent C++ FP64 certification run through v1.0.7 frozen protocol?

Must instantiate official v1.0.7 template and run native certification; not executed in this portable prototype.

- backend: Python FP64 reference only

H9 — PASS

Does exact pressure satisfy incompressible pressure-Poisson closure?

A046 analytical field control at two interior samples; no gravity metric claims.

- max_pressure_poisson_abs: 0.00027989862246613484

H10 — PASS

Does finite laboratory-frame kinetic-energy quadrature converge?

A057 reference energy control only; not at matched fixed circulation or topology and not a mass ratio.

- max_order36_60_relative_error: 6.544407857652665e-15

Backend provenance

{"cpp_certification": {"status": "NOT_RUN"}, "framework_target": "v1.0.7 native certification NOT RUN", "python_reference": {"authority": "REFERENCE", "precision": "FP64"}, "schema": "SST-HICKS-PROTOTYPE-BACKENDS"}

Report provenance

Declared ledger-to-summary digest binding: True
Files read: BACKEND_MANIFEST.json, GATE_LEDGER.json, RUN_SUMMARY.json
Scope: Read-only digest inventory and declared ledger/summary binding check only. Does not independently certify framework protocol, scientific results, or frozen output tree.